Enter the start and end date/time of your exam:
Sign the honor code pledge below before submitting your exam.

Start Date & Time: _____

End Date & Time: _____

ASEN 5090 Introduction to GNSS

FINAL EXAM

Due December 11, 2022 4 pm, in Gradescope

Format – There are two problems worth a total of 100 points.

You may write directly on the exam sheet and/or use your own paper or tablet computer to show your work. Please number all pages and scan them into a single pdf document for upload to Gradescope. All code used must be included in your submitted pdf.

The name of your pdf file should be: *LastName_FirstName_ASEN5090Final.pdf*

Logistics – The exam period begins on Friday, December 9, 2022 at 7 AM and close on Sunday, December 11, 2022 at 4PM. If you have a question about the exam you may email or text (303-884-1297 include your name) the instructor who will respond promptly to questions between 11am-noon on Friday, Saturday, or Sunday, and with some delay at other times during the exam period. Once you download the exam from Canvas, you have up to 24 hours to complete the exam, scan, and upload a pdf of your solutions to Gradescope. However, all solutions must be submitted by the end of the exam period. If you run into difficulty submitting your exam on Gradescope, you may email your solution file to the instructor.

Open Book – During the exam, students may use textbooks, class notes, and other references recommended for this course or that you collected/assembled prior to the date of the exam. The only internet site you may access once you begin the exam is our Canvas page. You may use a calculator or computer codes in solving the problems; however, extensive coding or typesetting is not expected.

During the entire exam period (7 AM Dec 9 to 4PM on Dec 11) you may NOT work with, consult, collaborate, or discuss the exam with ANYONE except the instructor or TA. Once you have downloaded the exam you may NOT access or search the web, online forums, or resources other than those listed above. Any violation of these exam conditions will result in a grade of F in the course and an honor code violation report will be filed with the honor code office.

If doing calculations by hand, please assume that the speed of light, $c = 3 \times 10^8$ m/s.

On my honor, as a University of Colorado Boulder student, I have followed all of the above instructions and have neither given nor received unauthorized assistance on this work.

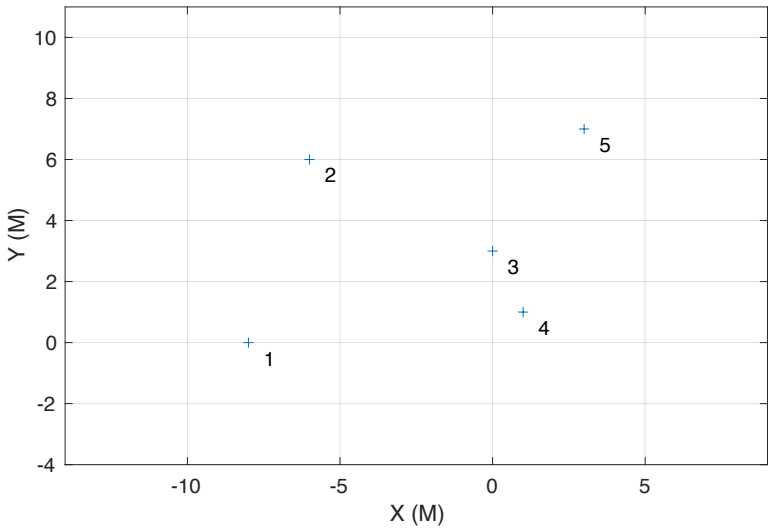
Signature Date

Printed name

PROBLEM 1. (50 points, a-f)

Consider a miniature two dimensional GPS-like system in which the receiver and all transmitters are located in the plane of the paper within the area shown in the figure. The table and figure below give the locations of 5 transmitters and the corresponding pseudorange measurements made by a receiver at an unknown location also within the area shown in the figure. Assume that all measurements have the same quality and that the only systematic error source is the receiver clock.

PRN	Satellite Position (m)	Pseudorange (m)
1	X1 = -8.0 Y1 = 0.0	17.0
2	X2 = -6.0 Y2 = 6.0	15.5
3	X3 = 0.0 Y3 = 3.0	15.9
4	X4 = 1. 0 Y4 = 1.0	17.4
5	X5 = 3.0 Y5 = 7.0	20.2



a. Select a reasonable initial guess for the receiver location and clock bias, and explain your reasoning.

b. Using the given information and your initial guess, set up a least squares solution to estimate the receiver location and briefly describe the procedure for solving it. Give the initial measurement sensitivity matrix (A matrix) and prefit residuals.

c. Solve for the receiver location and clock bias. Is it necessary to iterate? Does your solution converge? If so, how may iterations are required?

d. How important is it to have an accurate initial guess in this situation compared to the regular GPS geometry? Why?

e. Compute the HDOP using all 5 transmitters (based on your final result).

f. Choose what you think would be a particularly good or bad subset of three transmitters and recompute a new solution and HDOP. Explain the reasoning for your choice and whether your results make sense compared to the 5 measurement case above.

PROBLEM 2. (50 points, a-h)

This question deals with a simplified version of the GPS L5 civil signal. Assume the following:

Carrier Frequency: fcarrier = 1176.45 MHz

Code chip rate: fcode = 10.23 MHz, Full code length = 10,230 chips (or bits) (Call this C5)

Please assume that the speed of light c = $3x10^8$ m/s.

a) What is the duration of 1 code chip in seconds? In meters?

b) What is the duration of the entire simplified L5 code in seconds? In meters?

c) Sketch the overall spectrum of this simplified L5 Code, carefully identifying the center frequency, main lobe, and side lobe widths. How does it compare with the L1 C/A code? (*Note: The L5 code is actually designed differently from C/A so that it does not exhibit the 1kHz spectral lines we see with C/A code; so no need to worry about that aspect here.*)

d) Ignoring atmospheric losses, how much power must be transmitted by the satellite such that the signal power received by an antenna with Gain = 3 dB, located 20,000 km away is -155 dBW? Assume that the transmitting satellite antenna gain is 12 dB. You may ignore any other losses in the transmitter or receiver.

e) Compute the standard deviation of a C5 code DLL tracking error due to noise, for a signal with C/No = 45 dB-Hz a coherent integration time of 5 ms, and an early-late correlator spacing of 1 chip? How does this error for tracking C5 compare to a similarly configured tracking loop tracking a legacy C/A code signal with the same C/No.

f) How do the ionospheric delays for L5 compare to L1? Compute the L5 ionospheric delay that would be observed when the L1 ionospheric delay is 7 m.

g) How does the acquisition search for the simplified L5 code compare to L1 C/A? Comment on both the delay and Doppler ranges that must be considered.

h) Give 2 reasons that the L5 signals were added to GPS.